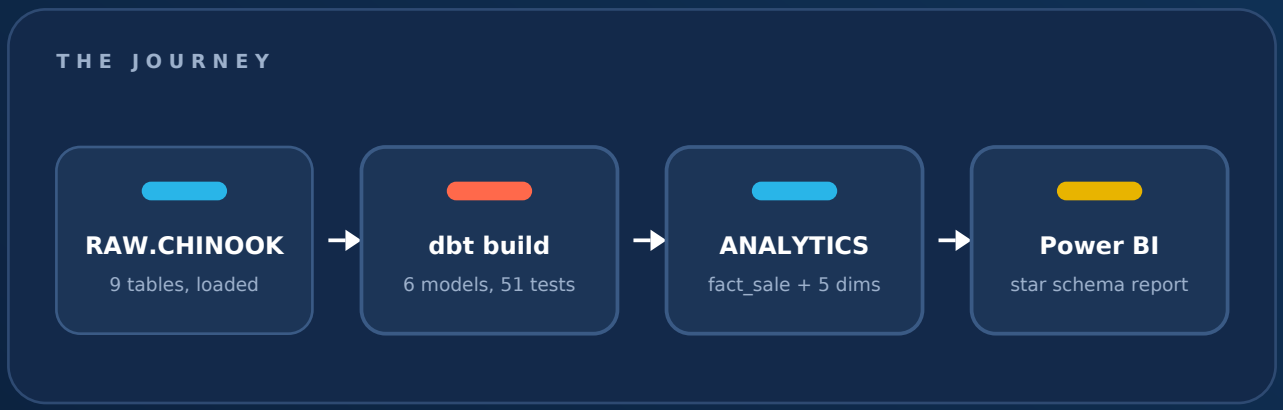
FIELD GUIDE 2 · MODEL & REPORT

Build the Music Sales star schema and connect Power BI

Turn RAW.CHINOOK into fact_sale and five dimensions with dbt, check every row, and load the star schema into Power BI.



Snowsight · dbt Studio IDE · Power BI Desktop · One folder to upload

Checked against docs.getdbt.com, docs.snowflake.com and learn.microsoft.com · 29 October 2026
Version 4 · a new music_sales project folder in ist722dbt, next to your assignment folders

Before you begin

This guide picks up where you are: RAW.CHINOOK is loaded and verified, SYSADMIN has USAGE and CREATE SCHEMA on ANALYTICS, your GitHub account is linked to dbt, and your dbt project uses your public repository `DarrelRamasray/ist722dbt` with a key pair from an earlier assignment.

The plan

Your repository already follows a pattern: each assignment has its own dbt project folder, and your one dbt project points at one of them. Music Sales becomes a fourth folder of the same kind.

FOLDER IN IST722DBT	WHAT IT IS	THIS GUIDE
`demo/`	Assignment dbt project	Left unchanged
`northwind/`	Assignment dbt project	Left unchanged
`quickstart/`	Assignment dbt project	Left unchanged
`music_sales/`	**New:** the Music Sales dbt project plus every script for it	You upload it in Part 4 and point dbt at it in Part 5

WHY A SEPARATE FOLDER, AND WHY ITS OWN SCHEMA

A separate folder keeps the Music Sales models out of your assignment projects, so neither can break the other. The Free Developer plan has one dbt project, so dbt works on one folder at a time through its **Project subdirectory** setting; you switch it to music_sales here and can switch it back whenever an assignment needs it (see **Switching between folders** after Part 9). The Music Sales tables also build in their own schema, your dbt schema plus `_CHINOOK`, so they can't replace tables such as DIM_CUSTOMER that an assignment built in your usual schema.

What you will build

TABLE	ROWS	BUILT FROM (RAW.CHINOOK)	DESIGN SHEET
`FACT_SALE`	2,240	INVOICELINE, INVOICE, CUSTOMER	fact_sale
`DIM_DATE`	1,817	INVOICE (first to last invoice date)	dim_date
`DIM_EMPLOYEE`	8	EMPLOYEE	dim_employee
`DIM_CUSTOMER`	59	CUSTOMER	dim_customer
`DIM_INVOICE`	412	INVOICE	dim_invoice
`DIM_TRACK`	3,503	TRACK, ALBUM, ARTIST, GENRE, MEDIATYPE	dim_track(2)

Columns, data types and tests come from `1. Bus_MatrixV7_simplified.xlsx`; the dbt files are the Snowflake versions of the sheets' SQL Server queries.

What is in the zip

Unzip it: right-click `Music_Sales_Next_Steps_v4.zip`, choose **Extract All**, and click **Extract**. Inside is this guide and the `music_sales` folder you will upload.

INSIDE MUSIC_SALES/	WHAT IT IS	USED IN
`dbt_project.yml`, `.gitignore`, `README.md`	Makes the folder a dbt project; builds every model as a table in the _CHINOOK schema	Parts 4–6

INSIDE MUSIC_SALES/	WHAT IT IS	USED IN
`models/chinook/` (8 files)	Sources, the six models and their 51 tests	Parts 4–6
`snowflake/05_check_dbt_role.sql`	Checks what dbt's role can reach	Part 2
`key_pair/key_fingerprint.sh`	Checks a private key file against Snowflake	Part 3
`key_pair/make_key_pair.sh`, `snowflake/06_set_public_key.sql`	A new key for Power BI, only if needed	Part 3
`snowflake/07_verify_star_schema.sql`	23 checks of the built tables	Part 7
`powerbi/music_sales_measures.dax`	Five Power BI measures	Part 9
`raw_staging_scripts/` (5 files)	The staging guide's scripts, kept with the project	Kept for reference

COPY SCRIPTS FROM THE FILES, NOT FROM THIS PDF

Copying from a PDF often joins every line into one. Open the file in VS Code or Notepad, press **Ctrl** + **A** then **Ctrl** + **C** , and paste.

CHECKPOINT

- ✓ The zip is unzipped.
- ✓ You can sign in to Snowflake, dbt and GitHub.
- ✓ A Windows computer for Power BI Desktop, which runs only on Windows.
- ✓ About ninety minutes.

PART 1 · IN THE DBT PLATFORM (BROWSER)

1 Check your dbt project

Write down seven settings. Change nothing yet.

WHERE: DBT PLATFORM

1 Open the project

In dbt, click your account name in the left menu, choose **Account settings**, then **Projects**, and click your project.

2 Note the repository and folder

The **Repository** should read `DarrelRamasray/ist722dbt`. Note the **Project subdirectory** value, for example `northwind`. You need it to switch back after this project.

3 Note the connection

Click the project's connection (or **Account settings › Connections** and open the Snowflake connection). Note **Account**, **Database**, **Warehouse** and **Role**.

4 Note your credentials

Click your account name, then **Your profile** › **Credentials**, and select the project. Note **Auth method** and **Schema**. If this page also shows a Role, Warehouse or Database, use those values instead of the connection's, because they override it.

5 Compare the account

In Snowsight, click your name (bottom left), then **Connect a tool to Snowflake**. The dbt Account must match this account, either as the **Account Identifier** (like `myorg-myaccount`) or as the account locator in the same dialog.

Write these down

SETTING	FROM	EXAMPLE	USED IN
Project subdirectory today	Step 2	`northwind`	Switching back after Part 9
Database	Step 3 or 4	`ANALYTICS`	Parts 2, 7, 8
Warehouse	Step 3 or 4	`COMPUTE_WH`	Parts 2, 7, 8
Role	Step 3 or 4	`SYSADMIN`	Parts 2, 7, 8
Schema	Step 4	`DBT_DRAMASRAY`	The tables land in this plus _CHINOOK
Auth method	Step 4	**Key pair**	Part 3
Account identifier	Step 5	`myorg-myaccount`	Part 8 (Power BI server)

STOP IF EITHER OF THESE IS TRUE

The Account doesn't match your Snowflake account: dbt would look for RAW.CHINOOK in the wrong place, so tell me before going on. Auth method is not Key pair: tell me too, because dbt no longer saves password credentials and the fix changes Part 3.

CHECKPOINT

- ✓ You have the seven values written down.
- ✓ The Account matches the Snowflake account that holds RAW.CHINOOK, and Auth method is Key pair.

PART 2 · IN SNOWSIGHT (BROWSER)

2 Check what dbt's role can reach

The Role from Part 1 must read RAW.CHINOOK and create a schema in the Database.

WHERE: SNOWSIGHT · APP.SNOWFLAKE.COM

1 Open a new SQL file

Projects › Workspaces › + Add New › SQL File , named `05_check_dbt_role.sql`. Paste in `music_sales/snowflake/05_check_dbt_role.sql` from the zip.

2 Fill in your values

Replace every `DBT_ROLE` with the Role from Part 1. If your Warehouse or Database differs from `COMPUTE_WH` or `ANALYTICS`, replace those too.

3 Run STEP 1, 2 and 3 one statement at a time

Click inside each statement and press **Ctrl** + **Enter** . Then use the table to decide about STEP 4.

YOUR ROLE	WHAT TO EXPECT	STEP 4?
`SYSADMIN` with Database `ANALYTICS`	STEP 2 returns 2240; STEP 3 shows the USAGE and CREATE SCHEMA rows you saw on 29 October	Skip
`ACCOUNTADMIN`	STEP 2 returns 2240 (ACCOUNTADMIN includes SYSADMIN's access); STEP 3 shows OWNERSHIP for ACCOUNTADMIN	Skip
Any other role	STEP 2 may fail with "does not exist or not authorized"	Run it, then run STEP 2 and 3 again

music_sales/snowflake/05_check_dbt_role.sql RUN IN SNOWSIGHT

```
-- ============================================================================
-- 05_check_dbt_role.sql
-- Run in: Snowsight > Projects > Workspaces, ONE step at a time
--         (click inside a statement, Ctrl+Enter, Mac: Cmd+Return)
-- What it does: checks that the role your dbt project uses can read
--   RAW.CHINOOK and can create a schema in the database dbt builds into.
--
-- Before running, replace these with the values you wrote down in Part 1:
--   DBT_ROLE    the Role dbt uses        (for example SYSADMIN)
--   COMPUTE_WH  the Warehouse dbt uses   (often COMPUTE_WH, then leave it)
--   ANALYTICS   the Database dbt uses    (often ANALYTICS, then leave it)
-- ============================================================================
-- ----------------------------------------------------------------------------
-- STEP 1. Switch to dbt's role and warehouse.
-- ----------------------------------------------------------------------------
USE ROLE DBT_ROLE;
USE WAREHOUSE COMPUTE_WH;
-- ----------------------------------------------------------------------------
-- STEP 2. Can the role read RAW.CHINOOK?
--   Expect 2240. An error means it can't (see STEP 4).
-- ----------------------------------------------------------------------------
SELECT COUNT(*) AS invoiceline_rows FROM RAW.CHINOOK.INVOICELINE;
-- ----------------------------------------------------------------------------
-- STEP 3. Can the role create a schema in dbt's database?
--   Look for a row whose grantee_name is your dbt role with OWNERSHIP,
--   or two rows with USAGE and CREATE SCHEMA.
--   If your dbt role is ACCOUNTADMIN and ACCOUNTADMIN owns the database,
--   that is also fine.
-- ----------------------------------------------------------------------------
SHOW GRANTS ON DATABASE ANALYTICS;
-- ----------------------------------------------------------------------------
-- STEP 4. ONLY if STEP 2 failed or STEP 3 showed nothing for your role:
--   run these lines (ACCOUNTADMIN can grant on objects it doesn't own).
--   Skip this step if STEP 2 returned 2240 and STEP 3 looked right.
-- ----------------------------------------------------------------------------
USE ROLE ACCOUNTADMIN;
GRANT USAGE ON DATABASE RAW TO ROLE DBT_ROLE;
GRANT USAGE ON SCHEMA RAW.CHINOOK TO ROLE DBT_ROLE;
GRANT SELECT ON ALL TABLES IN SCHEMA RAW.CHINOOK TO ROLE DBT_ROLE;
GRANT USAGE, CREATE SCHEMA ON DATABASE ANALYTICS TO ROLE DBT_ROLE;
```

CHECKPOINT

- ✓ STEP 2 returns 2240 using dbt's role.
- ✓ STEP 3 shows OWNERSHIP, or USAGE and CREATE SCHEMA, for that role.

PART 3 · ON YOUR COMPUTER, THEN SNOWSIGHT

3 Find the private key for Power BI

dbt already has your key. Power BI needs the key file itself.

WHERE: GIT BASH + SNOWSIGHT

dbt already signs in with your key, but Power BI asks for the key file itself, and dbt isn't a reliable place to copy it back from. Find the file and confirm it is the one Snowflake knows. If you can't, 3C adds a second key just for Power BI and leaves dbt untouched.

3A · Find the file

1 Search for it

In File Explorer, click **This PC**, type `*.p8` in the search box at the top right, and press **Enter** . Also check the folder you used for the earlier assignment and `C:\Users\<you>\.ssh`. Private keys usually end in `.p8` or `.pem`.

2 Note its full path

Right-click the file and choose **Copy as path**. It looks like `"C:\Users\darre\snowflake_keys\rsa_key.p8"`. If you find no file, skip to 3C.

3B · Check it matches Snowflake

3 Read Snowflake's fingerprint

In Snowsight, run `DESC USER <username>;` with your username from `SELECT CURRENT_USER();`. Copy the value of `RSA_PUBLIC_KEY_FP`; it starts with `SHA256:`.

4 Open Git Bash in the key_pair folder

In File Explorer, open the unzipped `music_sales\key_pair` folder, right-click an empty part of it, and choose **Open Git Bash here** (Windows 11: **Show more options** first). If Git Bash isn't installed, get it from `git-scm.com` › **Download for Windows** and install with the defaults.

5 Print the file's fingerprint

Type the line below with your path: change `C:\` to `/c/` and every backslash to a forward slash, and keep the double quotes.

Git Bash — TYPE, THEN PRESS ENTER

```
bash key_fingerprint.sh "/c/Users/darre/snowflake_keys/rsa_key.p8"
```

WHAT YOU SEE	WHAT IT MEANS
The fingerprint equals RSA_PUBLIC_KEY_FP (ignoring `SHA256:`)	This is your key. Use this file in Part 8 and skip 3C
First line is `-----BEGIN ENCRYPTED PRIVATE KEY-----`	The key has a passphrase. openssl asks for it now, and Power BI will ask for it too
The fingerprints differ, or no file was found	Do 3C

3C · Only if needed: add a second key for Power BI

6 **Make the key pair**

In the same Git Bash window, run `bash make_key_pair.sh`. It saves `rsa_key.p8` and two public-key files in `C:\Users\<you>\snowflake_keys` and prints the public key on one line. It stops without overwriting anything if that folder already has a key.

7 **Attach it in slot 2**

In Snowsight, open a new SQL file named `06_set_public_key.sql`, paste in `music_sales/snowflake/06_set_public_key.sql`, and run its four steps one at a time. It uses `RSA_PUBLIC_KEY_2`, so the key dbt uses in slot 1 keeps working.

CHECKPOINT

- ✓ You know the full path of a private key file whose fingerprint matches RSA_PUBLIC_KEY_FP, or RSA_PUBLIC_KEY_2_FP after 3C.
- ✓ You know whether it has a passphrase.

KEEP THE KEY OUT OF THE REPOSITORY

The key file stays in its own folder on your computer. Don’t copy it into music_sales: ist722dbt is public, so anyone can read what is in it.

PART 4 · IN YOUR BROWSER

4 Upload the music_sales folder

One upload puts the whole Music Sales project into ist722dbt.

WHERE: GITHUB.COM

1 **Open the repository**

Go to `github.com/DarrelRamasray/ist722dbt`. The branch menu above the file list must read `main`.

2 **Start an upload**

Click **Add file › Upload files** .

3 **Drag in the folder**

In File Explorer, open the unzipped `Music_Sales_Next_Steps_v4` folder and drag the **music_sales** folder (the folder itself, not its contents) onto the upload area. GitHub lists every file with its path, starting with `music_sales/`.

4 **Commit**

Under **Commit changes**, type `Add Music Sales dbt project`, leave **Commit directly to the main branch** selected, and click **Commit changes**.

5 **Check what arrived**

Click the new `music_sales` folder. It should show `key_pair`, `models`, `powerbi`, `raw_staging_scripts`, `snowflake`, `.gitignore`, `README.md` and `dbt_project.yml`.

IF .GITIGNORE IS MISSING

Some browsers skip files whose names start with a dot. Inside music_sales on GitHub, click **Add file › Create new file** , name it `.gitignore`, paste the three lines below, and click **Commit changes**. It stops dbt's working files from showing up as changes.

music_sales/.gitignore — CREATE ON GITHUB IF MISSING

```
target/
dbt_packages/
logs/
```

CHECKPOINT

- ✓ ist722dbt shows music_sales next to demo, northwind and quickstart.
- ✓ music_sales contains dbt_project.yml, .gitignore, README.md and the five folders.

PART 5 · IN THE DBT PLATFORM, THEN STUDIO IDE

5 Point dbt at music_sales

Switch the project's folder, open the Studio IDE, and confirm dbt sees the six models.

WHERE: DBT PLATFORM + STUDIO IDE

1 Change the Project subdirectory

Go to **Account settings › Projects** , click your project, then **Edit**. In **Project subdirectory**, replace the folder name from Part 1 with `music_sales` (exactly, lower case). Click **Save**.

2 Open the Studio IDE

Click **Studio** in the left menu. If it was already open, restart it so it reads the new folder: click the three dots next to the IDE status at the bottom right and choose **Restart IDE**.

3 Get the latest main

If the branch shown by the git button isn't `main`, click **Change branch** and choose `main`. If the git button says **Commit and sync** before you switch, you have uncommitted work from an assignment on that branch: commit it there first. Then click **Pull from remote**.

4 Confirm dbt sees the project

In the command bar at the bottom, run the line below. It should list six models, each starting with `music_sales.chinook.`, for example `music_sales.chinook.fact_sale`.

Studio IDE command bar — TYPE, THEN PRESS ENTER

```
dbt ls --resource-type model
```

WHAT DBT NOW USES

With the subdirectory set to music_sales, dbt reads only that folder's dbt_project.yml and models. Your assignment folders are still in the repository, untouched; dbt just isn't pointed at them.

CHECKPOINT

- ✓ Project subdirectory reads music_sales.
- ✓ dbt ls lists six models starting with music_sales.chinook.

PART 6 · IN THE DBT STUDIO IDE

6 Build and test the star schema

One command builds the six tables and runs the 51 tests.

WHERE: DBT STUDIO IDE COMMAND BAR

1 Run the build

In the command bar, type the line below and press **Enter** . Because the project holds only the Music Sales models, a plain build is enough.

Studio IDE command bar — TYPE, THEN PRESS ENTER

```
dbt build
```

2 Wait for it to finish

dbt builds each model and runs its tests. The relationships tests on fact_sale wait until the five dimensions exist.

3 Read the summary

It should report 57 nodes with no errors: 6 models and 51 tests. The wording differs slightly between dbt versions.

4 If anything is red

Click the failed model or test to read its log, then look up the message in Troubleshooting.

WHERE THE TABLES GO

dbt creates the schema the first time: your Schema from Part 1 plus `_CHINOOK`, in the Database from Part 1, for example `ANALYTICS.DBT_DRAMASRAY_CHINOOK`. That needs the CREATE SCHEMA permission checked in Part 2.

CHECKPOINT

- ✓ The run finished with 57 successes and 0 errors.

PART 7 · IN SNOWSIGHT (BROWSER)

7 Verify the star schema

Twenty-three checks compare what dbt built with the source data.

WHERE: SNOWSIGHT

1 Open a new SQL file

Projects › Workspaces › + Add New › SQL File , named `07_verify_star_schema.sql`. Paste in `music_sales/snowflake/07_verify_star_schema.sql`.

2 Run STEP 1

If your Role from Part 1 isn't SYSADMIN, change `USE ROLE SYSADMIN;` first. Run the two lines and note **database_name** and **schema_name**. If several rows appear, for example because an assignment also built a FACT_SALE, use the one whose schema ends in CHINOOK.

3 Edit and run STEP 2

Replace `ANALYTICS.YOUR_DBT_SCHEMA` with database_name.schema_name from STEP 1, and the warehouse if yours differs. Run both lines.

4 Run STEP 3

Click inside the `WITH checks AS` query and press **Ctrl** + **Enter** . A grid of 23 rows appears.

CHECKS	WHAT THEY PROVE
1–6	Each table has the expected number of rows
7–8	Sales total $2,328.60 and 2,240 units, the same as RAW.CHINOOK
9–14	Every FACT_SALE key points at a row in its dimension, and no sale is missing its employee
15–20	DIM_DATE covers every invoice date, runs 2009-01-01 to 2013-12-22 without gaps, and names weekdays correctly
21	DIM_TRACK's joins work (track 1 is by AC/DC)
22–23	SALEAMOUNT is NUMBER(21,2) and INVOICEDATEKEY is DATE, as in the design

CHECKPOINT

- ✓ All 23 rows say PASS.
- ✓ You wrote down the database and schema for Power BI, e.g. ANALYTICS.DBT_DRAMASRAY_CHINOOK.

PART 8 · IN POWER BI DESKTOP (WINDOWS)

8 Connect Power BI to the star schema

Sign in with the key file from Part 3 and import the six tables.

WHERE: POWER BI DESKTOP

1 Install or update Power BI Desktop

Open the Microsoft Store, search for `Power BI Desktop`, and click **Get** (or update it under **Library**; key-pair sign-in needs a recent version).

2 Start a blank report

Open Power BI Desktop and choose **Blank report**.

3 Turn off automatic relationships

Click **File › Options and settings › Options** . Under **CURRENT FILE**, click **Data Load**, untick **Autodetect new relationships after data is loaded**, and click **OK**. Columns such as INVOICEID appear in more than one table, and autodetect could link the wrong ones.

4 Open the Snowflake connector

Click **Home › Get data › More…** , select **Database**, select **Snowflake**, and click **Connect**.

5 Fill in the connection

BOX	ENTER
Server	Your account identifier from Part 1 plus `.snowflakecomputing.com`, e.g. `myorg-myaccount.snowflakecomputing.com`
Warehouse	The Warehouse from Part 1, e.g. `COMPUTE_WH`
Advanced options › Role name	The Role from Part 1, e.g. `SYSADMIN`. It owns the tables dbt built, so it can read them
Advanced options › Database	The database from Part 7, e.g. `ANALYTICS`
Data Connectivity mode	**Import** (2,240 fact rows)

6 Sign in with the key pair

Click **OK**. In the sign-in window, choose the key-pair option on the left (labelled Key Pair or KeyPair), enter your Snowflake username, and browse to the key file from Part 3. If the key has a passphrase, a box appears for it. Click **Connect**.

7 Pick the six tables

In **Navigator**, expand the database and then the schema from Part 7 (e.g. `DBT_DRAMASRAY_CHINOOK`). Tick DIM_CUSTOMER, DIM_DATE, DIM_EMPLOYEE, DIM_INVOICE, DIM_TRACK and FACT_SALE, then click **Load**.

8 Save the file

Click **File › Save as** and save it as `MusicSales.pbix` somewhere outside the music_sales folder, for example in Documents.

NO KEY-PAIR OPTION IN THE SIGN-IN WINDOW?
Update Power BI Desktop first. If it still only offers username/password and Microsoft account, Snowflake's documentation lists Power BI among the apps where a programmatic access token can replace a password; by default a token needs a network policy on your user and expires after 15 days, so ask your instructor before using one.

CHECKPOINT
- ✓ The Data pane lists DIM_CUSTOMER, DIM_DATE, DIM_EMPLOYEE, DIM_INVOICE, DIM_TRACK and FACT_SALE.
- ✓ MusicSales.pbix is saved.

PART 9 · IN POWER BI DESKTOP

9

Model the star schema

Five relationships, a date table, sort orders, five measures and a quick test.

WHERE: POWER BI DESKTOP

9A · Create the five relationships

1 Open Manage relationships

Click the **Model view** icon on the left edge, then **Home › Manage relationships** . If any relationships are listed, select each and click **Delete**.

2 Add each relationship in the table

Click **New relationship**, choose the tables and columns, set **Cardinality** to **Many to one (*:1)** and **Cross-filter direction** to **Single**, leave **Make this relationship active** ticked, and click **Save** (or OK). Repeat five times.

FROM TABLE	FROM COLUMN	TO TABLE	TO COLUMN
FACT_SALE	INVOICEKEY	DIM_INVOICE	INVOICEKEY
FACT_SALE	TRACKKEY	DIM_TRACK	TRACKKEY
FACT_SALE	CUSTOMERKEY	DIM_CUSTOMER	CUSTOMERKEY
FACT_SALE	EMPLOYEEKEY	DIM_EMPLOYEE	EMPLOYEEKEY
FACT_SALE	INVOICEDATEKEY	DIM_DATE	DATEKEY

These are the five FK Ref Dimension.col entries in the fact_sale design sheet. Don't add relationships between dimensions.

9B · Mark DIM_DATE as the date table

3 Mark it

Click the **Table view** icon, select **DIM_DATE** in the Data pane, click **Table tools › Mark as date table** , choose **DATEKEY**, and confirm. It is accepted because DATEKEY has one row per day with no gaps (check 18 in Part 7).

9C · Sort the name columns

4 Sort month and day names

In Table view on DIM_DATE: click the **MONTHNAME** column, then **Column tools › Sort by column › MONTH** . Click **DAYNAME**, then **Column tools › Sort by column › DAYOFWEEK** .

5 Stop number columns being added up

For YEAR, QUARTER, MONTH, DAYOFMONTH and DAYOFWEEK: click the column and set **Column tools › Summarization › Don't summarize** .

9D · Add the measures

6 Add each measure

Click the **Report view** icon, select **FACT_SALE** in the Data pane, and click **Home › New measure** . Copy one measure line from `music_sales/powerbi/music_sales_measures.dax` into the formula bar and press **Enter** . Repeat for all five.

music_sales/powerbi/music_sales_measures.dax — POWER BI DESKTOP

```
// ====================================================================
// music_sales_measures.dax
// Use in: Power BI Desktop, guide Part 9, step "Add the measures".
// For each measure: select the FACT_SALE table, click New measure,
// then copy ONE line below (the line that starts with the measure name)
// into the formula bar and press Enter.
// Expected value with no filters is in the comment above each line.
// ====================================================================
// Expected: 2,328.60
Total Sales = SUM ( FACT_SALE[SALEAMOUNT] )
// Expected: 2,240
Units Sold = SUM ( FACT_SALE[QUANTITY] )
// Expected: 1.04  (unit price is non-additive, so it is averaged, not summed)
Average Unit Price = AVERAGE ( FACT_SALE[UNITPRICE] )
// Expected: 412
Invoice Count = DISTINCTCOUNT ( FACT_SALE[INVOICEID] )
// Expected: 59
Customers With Sales = DISTINCTCOUNT ( FACT_SALE[CUSTOMERKEY] )
```

9E · Test the model

7 Build two quick visuals

Add a **Card** with Total Sales: it shows 2,328.60. Add a **Clustered column chart** with DIM_DATE[YEAR] on the X-axis and Total Sales on the Y-axis, and compare with the table.

YEAR	2009	2010	2011	2012	2013	TOTAL
Total Sales ($)	449.46	481.45	469.58	477.53	450.58	2,328.60

Optional third check: DIM_EMPLOYEE[FIRSTNAME] with Total Sales shows Jane 833.04, Margaret 775.40 and Steve 720.16. These numbers come from the data you loaded.

CHECKPOINT

- ✓ Model view shows exactly five relationships, all from FACT_SALE.
- ✓ DIM_DATE is marked as the date table.
- ✓ The card shows 2,328.60 and the yearly columns match the table.

AFTER PART 9 · DBT PLATFORM

Switching between folders

Point dbt back at an assignment folder, or back at music_sales.

WHERE: DBT PLATFORM

Your one dbt project works on one folder at a time. The tables already built in Snowflake stay where they are whichever folder dbt points at, so Power BI keeps working after you switch.

1. **Change the folder**

   Go to **Account settings › Projects** , click your project, then **Edit**. Set **Project subdirectory** to the folder you want (for example `northwind`, the value from Part 1, or `music_sales`), and click **Save**.

2. **Restart the Studio IDE**

   Click the three dots next to the IDE status at the bottom right and choose **Restart IDE**. Run `dbt ls --resource-type model`; the names now start with that folder's project name.

BEFORE YOU SWITCH

Commit any changes in the Studio IDE first. If you set up a dbt deployment job for an assignment, it also uses the Project subdirectory, so it builds whichever folder is selected when it runs.

Troubleshooting

Find the message you see, then apply the fix.

YOU SEE	LIKELY CAUSE	FIX
05 STEP 2: `does not exist or not authorized`	dbt's role can't see RAW.CHINOOK	Run STEP 4, then STEP 2 again
`openssl: command not found`	You are in Command Prompt, not Git Bash	Open Git Bash (Part 3, step 4)
key_fingerprint.sh prints an error about the file	Path typed with C:\ or backslashes	Use /c/... with forward slashes, inside double quotes
ALTER USER: `Insufficient privileges`	Your role can't change users	Run `USE ROLE ACCOUNTADMIN;` then the ALTER USER line again
GitHub shows the files but no music_sales folder	The folder's contents were dragged instead of the folder	Delete the uploaded files and drag the music_sales folder itself (Part 4, step 3)
Studio IDE: `dbt_project.yml` not found, or no models listed	Project subdirectory misspelled, or the IDE didn't restart	Set it to exactly music_sales, save, then Restart IDE
Studio IDE: Commit and sync shows assignment files	Earlier uncommitted work	Commit it on its own branch before switching (Part 5, step 3)
dbt ls lists northwind or quickstart models	The subdirectory change didn't take effect	Check Part 5, step 1 saved, then Restart IDE
dbt: `Database 'RAW' does not exist or not authorized`	dbt's role can't see RAW	Part 2, STEP 4
dbt: `Insufficient privileges to operate on database`	dbt's role can't create the _CHINOOK schema	Part 2, STEP 4 (the CREATE SCHEMA line)
dbt: source `chinook` not found	_chinook__sources.yml missing or in the wrong folder	It must be in music_sales/models/chinook
dbt: a relationships test fails	A key has no match in its dimension	Run 07_verify_star_schema.sql; checks 9–13 name the key
07 STEP 1 shows several FACT_SALE rows	An assignment also built a FACT_SALE table	Use the row whose schema ends in CHINOOK
Power BI: `We couldn't authenticate with the credentials provided`	Username or key doesn't match	**File › Options and settings › Data source settings** , select the Snowflake source, **Clear Permissions**, and sign in again
Power BI: the six tables aren't in Navigator	Wrong Role, or the schema isn't expanded	Role must be dbt's role from Part 1; expand the schema from Part 7
Power BI: totals too high	An extra or wrong relationship	Model view: keep only the five in Part 9A
Power BI: YEAR shows as 2,011,000-style numbers	Summarization still on	Part 9, step 5
Power BI: months in alphabetical order	Sort by column not set	Part 9, step 4

Quick reference

TASK	WHERE	COMMAND OR CLICK PATH
Switch the project folder	dbt platform	Account settings › Projects › your project › Edit › Project subdirectory
Connection (account, database, role)	dbt platform	Account settings › Connections
Your credentials	dbt platform	Your profile › Credentials
Restart the Studio IDE	Studio IDE	Three dots next to the IDE status (bottom right) › Restart IDE
Check a key file's fingerprint	Git Bash in music_sales/key_pair	`bash key_fingerprint.sh "/c/.../file.p8"`
List the project's models	Studio IDE command bar	`dbt ls --resource-type model`
Build and test everything	Studio IDE command bar	`dbt build`
Rebuild one model and its tests	Studio IDE command bar	`dbt build --select fact_sale`
Find the built tables	Snowsight	`SHOW TABLES LIKE 'FACT_SALE' IN ACCOUNT;`
Refresh Power BI	Power BI Desktop	Home › Refresh

Recommended next steps

STEP	WHY
Present it at Milestone 2 (11/2)	The syllabus asks for the detail dimensional model and its implementation for at least one business process; Parts 4–7 are that implementation for Music Sales
Send your professor the link	github.com/DarrelRamasray/ist722dbt/tree/main/music_sales shows the whole project; the repository is public, and Class 07 asks for a repo your professor can access
Make later changes on a branch	In the Studio IDE: Create branch, edit, Commit and sync, then Create a pull request on GitHub and merge it. That keeps a record of each change
Set up a dbt deployment environment and job	So main builds on a schedule into a production schema. dbt doesn't support passwords for deployment credentials, so use a key pair there too. Remember the job follows the Project subdirectory
Build the report for Milestone 3 (11/30)	Use the Part 9 measures and the dimensions to answer the Music Sales functional requirements
Add more business processes here	New facts such as fact_rep_portfolio can go in music_sales/models/chinook and ref dim_employee and dim_date
Add incremental loads (Class 10)	When the source changes, reload RAW with the staging guide's Reloading section and rerun dbt build

Sources

- dbt: Connect Snowflake (docs.getdbt.com/docs/platform/connect-data-platform/connect-snowflake): key pair as the default, no new password credentials.
- dbt: Connect to GitHub (docs.getdbt.com/docs/platform/git/connect-github): Linked accounts, project repository settings, .gitignore entries, Restart IDE, Pull from remote.

- dbt: About the Studio IDE (docs.getdbt.com/docs/platform/studio-ide/develop-in-studio): Credentials page, Studio menu, command bar, Change branch.
- dbt: Add data tests (docs.getdbt.com/docs/build/data-tests): data_tests and the arguments property for relationships.
- dbt pricing (getdbt.com/pricing): the Developer plan includes one Developer seat and one project.
- Microsoft Learn: Power Query Snowflake connector (learn.microsoft.com/en-us/power-query/connectors/snowflake): KeyPair sign-in, passphrase prompt, Role and Database options, Import mode.
- Snowflake: Key-pair authentication (docs.snowflake.com/en/user-guide/key-pair-auth): RSA_PUBLIC_KEY and RSA_PUBLIC_KEY_2, fingerprints in DESC USER. Programmatic access tokens (docs.snowflake.com/en/user-guide/programmatic-access-tokens).
- IST 722 syllabus (Fall 2026): Milestones on 11/2 and 11/30. Class 07 slides: GitHub repo with professor access.

The dbt models, the 51 tests and the 23 Snowflake checks were tested against the same Chinook data in a local database, not in Snowflake or dbt; Parts 6 and 7 confirm them on your account.